



# Speech Anatomy 1

## Imaging

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## Seeing Speech

Maybe you think speech looks like this:

|                     | Bilabial | Labiodental | Dental | Alveolar | Postalveolar | Retroflex | Palatal | Velar | Uvular | Pharyngeal | Glottal |
|---------------------|----------|-------------|--------|----------|--------------|-----------|---------|-------|--------|------------|---------|
| Plosive             | p b      |             |        | t d      |              | ʈ ɖ       | c ɟ     | k ɡ   | q ɢ    |            | ʔ       |
| Nasal               | m        | ɱ           |        | n        |              | ɳ         | ɲ       | ŋ     | ɴ      |            |         |
| Trill               | ʙ        |             |        | r        |              |           |         |       | ʀ      |            |         |
| Tap or Flap         |          | ɸ           |        | ɾ        |              | ɽ         |         |       |        |            |         |
| Fricative           | ɸ β      | f v         | θ ð    | s z      | ʃ ʒ          | ʂ ʐ       | ç ʝ     | x ɣ   | χ ʁ    | ħ ʕ        | h ɦ     |
| Lateral fricative   |          |             |        | ɬ ɮ      |              |           |         |       |        |            |         |
| Approximant         | w        | ʋ           |        | ɹ        |              | ɻ         | j       | ɰ     |        |            |         |
| Lateral approximant |          |             |        | l        |              | ɭ         | ʎ       | ʟ     |        |            |         |

## Seeing speech

Speech is a physical activity

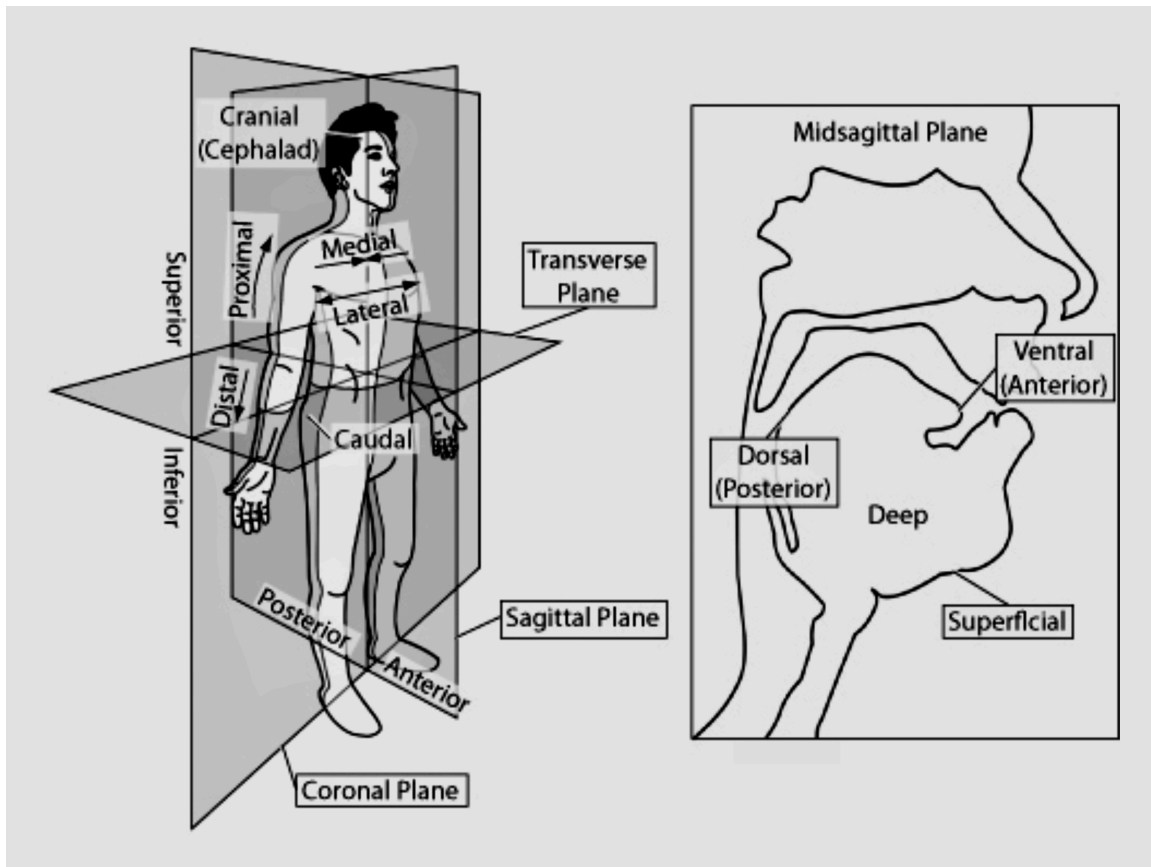
videos/xray.webm

## Today

- Tools we use for looking at speech.
- Starting some anatomy.

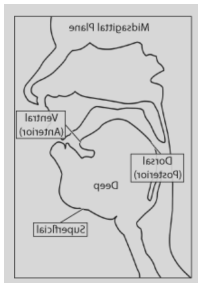


## Conceptualizing the head



Posterior    Anterior  
“back”      “front”

## Conceptualizing the head



|                     | Bilabial | Labiodental | Dental | Alveolar | Postalveolar | Retroflex | Palatal | Velar | Uvular | Pharyngeal | Glottal |
|---------------------|----------|-------------|--------|----------|--------------|-----------|---------|-------|--------|------------|---------|
| Plosive             | p b      |             |        | t d      |              | ʈ ɖ       | c ɟ     | k ɡ   | q ɢ    |            | ʔ       |
| Nasal               | m        | ɱ           |        | n        |              | ɳ         | ɲ       | ŋ     | ɴ      |            |         |
| Trill               |          |             |        | r        |              |           |         |       | ʀ      |            |         |
| Tap or Flap         |          | v           |        | ɾ        |              | ɽ         |         |       |        |            |         |
| Fricative           | ɸ β      | f v         | θ ð    | s z      | ʃ ʒ          | ʂ ʐ       | ç ʝ     | x ɣ   | χ ʁ    | ħ ʕ        | h ɦ     |
| Lateral fricative   |          |             |        | ɬ ɮ      |              |           |         |       |        |            |         |
| Approximant         | w        | ʋ           |        | ɹ        |              | ɻ         | j       | ɰ     |        |            |         |
| Lateral approximant |          |             |        | l        |              | ɭ         | ʎ       | ʟ     |        |            |         |

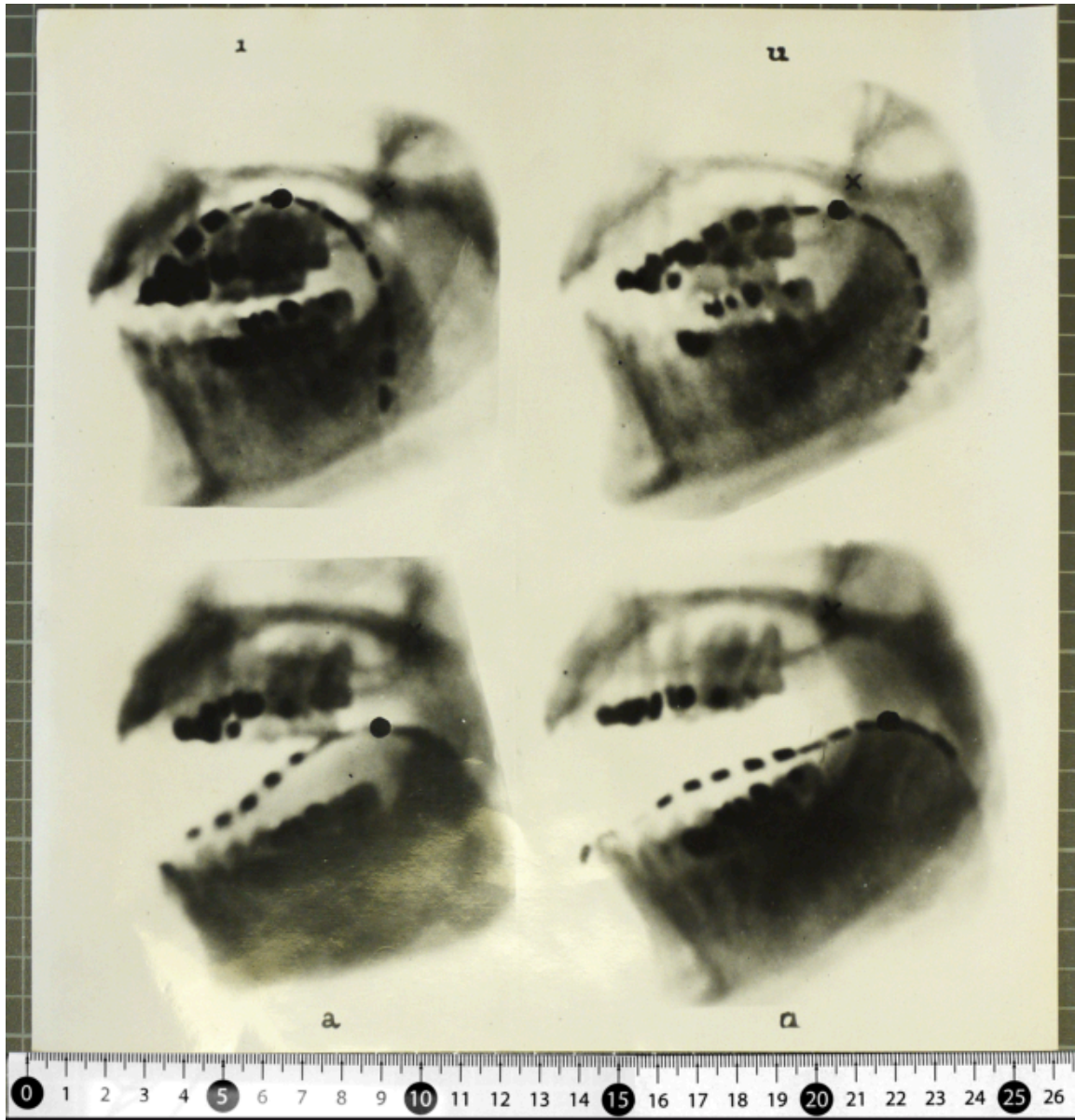


## Tools of the trade

### X-Ray

anterior ←

→ posterior





## Tools of the trade

### X-Ray

#### Pros?

- Better than nothing

#### Cons?

- Can't image fleshy bits well
- Harmful to health
- Static/low framerate

## Tools of the trade

### Ultrasound tongue imaging

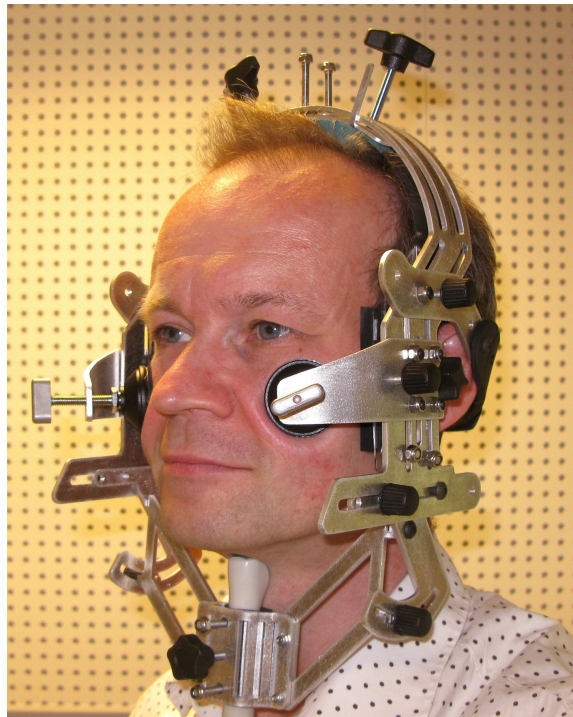


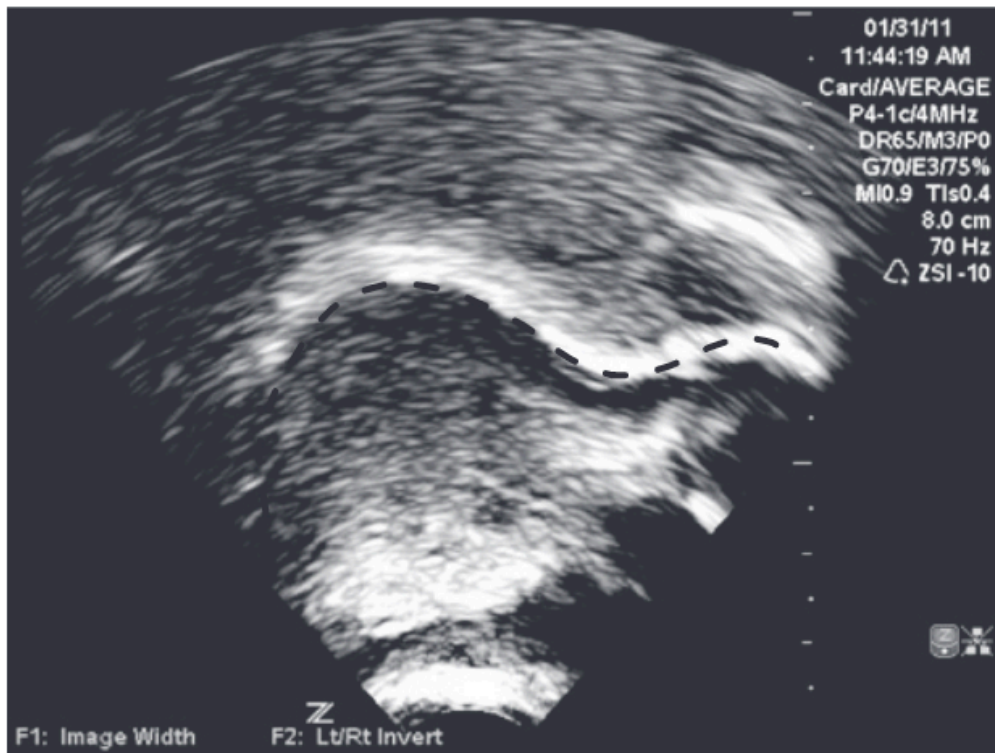
Figure 1: source: <https://www.articulateinstruments.com/ultrasound-imaging/?target=Micro%2FSonoSpeech>

## Tools of the trade

### Ultrasound tongue imaging

posterior ←

→ anterior



S. Lin, P. S. Beddor, and A. W. Coetzee [1]

## Tools of the trade

### Ultrasound tongue imaging

#### Pros?

- Relatively non-invasive.

#### Cons?

- Can *only* image fleshy bits.
- Can't image across an air gap.
- Relatively low frame rate.



## Tools of the trade

### Electromagnetic Articulography (EMA)

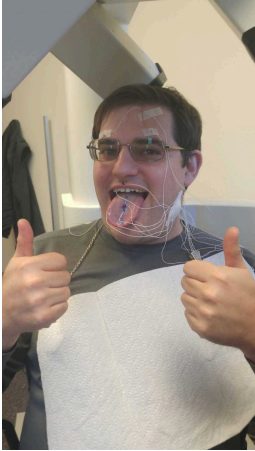
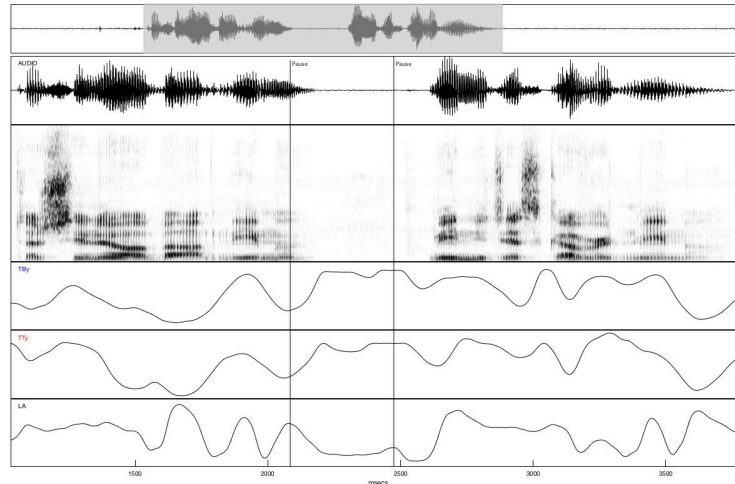


Figure 1: Credit:  
Will Styler



## Tools of the trade

### Electromagnetic Articulography (EMA)

#### Pros?

- Very high time resolution.
- Spatially precise.

#### Cons?

- Invasive.
- Harder to visualize.
- Expensive equipment.

## Tools of the trade

### MRI

posterior

anterior

videos/mri.mp4





## Tools of the trade

### MRI

#### Pros?

- Non invasive.
- High quality images.

#### Cons?

- Low frame rate (but improving).
- Very loud.
- Speakers must be lying down.
- Expensive equipment.

## Seeing Speech

**SEEING SPEECH**

- Introduction
- About the project
- IPA Charts
- How UTI works
- How MRI works
- Recording UTI and lips
- Recording MRI
- Comparing MRI and UTI
- Creating the animations
- The Phoneticians
- /j/ and /j/ in English
- Teangannan na Gàidhlig
- Phonetics in Scotland
- Contact us

### The International Phonetic Alphabet (revised to 2005)

There are two MRI viewing options for both charts: MRI 1 and MRI 2.

- **MRI 1** was recorded in 2014 at Edinburgh Imaging Facility Queen's Medical Research Institute (QMRI), with a frame rate of 7.5fps. Each file has been overdubbed with "clean" audio, recorded in a recording studio.
- **MRI 2** was recorded in 2022-2023 at the Edinburgh Imaging Facility, Royal Infirmary Edinburgh (RIE), with a frame rate of 6.25fps and a 70% increase in image resolution, compared to MRI 1. Audio for the MRI 2 files was recorded in the MRI machine using an optical microphone and post-hoc noise cancelling. We aim to replace the MRI 2 audio with clean audio in the near future.

Consonants (Pulmonic) Consonants (Non-Pulmonic) Vowels Other Symbols

#### Consonants (Pulmonic - produced with air from the lungs)

Video type: Animation MRI 1 MRI 2 Ultrasound Speaker: Janet Beck

Press on a symbol to play the associated video.

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| Nasal               | m        | ɱ           |        | n        |              | ɳ         | ɲ       | ŋ     | ɴ      |            |         |
| Trill               | ʙ        |             |        | r        |              |           |         |       | ʀ      |            |         |
| Tap or Flap         |          |             | ɾ      |          |              | ɽ         |         |       |        |            |         |
| Fricative           | ɸ β      | f v         | θ ð    | s z      | ʃ ʒ          | ʂ ʐ       | ç ʝ     | x ɣ   | χ ʁ    | ħ ʕ        | h ɦ     |
| Lateral fricative   |          |             |        | ɬ ɮ      |              |           |         |       |        |            |         |
| Approximant         |          |             | ʋ      | ɹ        |              | ɻ         | j       | ɰ     |        |            |         |
| Lateral approximant |          |             |        | l        |              | ɭ         | ʎ       | ʟ     |        |            |         |

## References

### Bibliography

- [1] S. Lin, P. S. Beddor, and A. W. Coetzee, "Gestural reduction , lexical frequency , and sound change : A study of post-vocalic / l / , " vol. 5, no. 1, pp. 9–36, 2014, doi: 10.1515/lp-2014-0002.